Introduction

In cluster-randomised trials (CRTs), treatment is assigned at the cluster (clinic, school, village) rather than individual level. Within-cluster correlation inflates variance, reducing effective sample size and requiring more individuals than an individually randomised trial.

Prerequisites

Intraclass correlation, CRT design.

Theory

The **design effect** (DE) converts individual-level sample size to cluster-RCT sample size:

$$\text{DE} = 1 + (\bar{m} - 1)\rho_{ICC},$$

where $\bar{m}$ is the average cluster size and ρ_{ICC} is the intraclass correlation.

Required individual-level n for an individually-randomised design, multiplied by DE, gives the cluster-RCT total. Required clusters = total / $\bar{m}$.

Assumptions

- Balanced or nearly balanced cluster sizes.
- Pre-specified ICC (from pilot or literature).
- Analysis adjusts for clustering (mixed models or GEE).

R Implementation

```r
library(clusterPower)

# Continuous outcome: detect a 5-unit mean difference, sigma = 15
# ICC = 0.02, 30 subjects per cluster
# Individual n for same power:
n_ind <- 2 * 15^2 * (qnorm(0.975) + qnorm(0.80))^2 / 5^2
n_ind

rho <- 0.02
m_bar <- 30
DE <- 1 + (m_bar - 1) * rho
c(design_effect = DE,
  cluster_total_n = n_ind * DE,
  clusters_per_arm = ceiling(n_ind * DE / m_bar / 2))

# Direct calculation via clusterPower
cpa.normal(alpha = 0.05, power = 0.80, nclusters = NA,
           nsubjects = 30, d = 5 / 15,   # d = delta / sigma
           ICC = 0.02)
```

Output & Results

Individual $n \approx 143$ per arm; DE = 1.58 at ICC 0.02 and cluster size 30, giving 226 per arm (about 8 clusters of 30 per arm).

Interpretation

"With an assumed ICC of 0.02 and 30 subjects per cluster, the design effect is 1.58. To detect a standardised difference of 0.33 (5 units / 15 SD) at 80 % power and two-sided $\alpha = 0.05$, 8 clusters per arm are required (total 480 subjects)."

Practical Tips

- ICC estimates vary widely; use pilot data, literature, or sensitivity ranges (0.001 to 0.1).
- Unequal cluster sizes reduce power; inflate n by 10-20 % as buffer.
- Large m increases DE linearly in ICC; adding clusters is more efficient than more per-cluster recruitment.
- Stratified or matched CRTs (pairs of clusters) reduce DE.
- Analysis model: linear mixed model with random cluster effects; GEE with exchangeable correlation; cluster-level summary.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Sample size, power, and Quarto reporting
- Power: closed-form and simulation